Home Work # 3

1. Linear design.

(i) fiber traverse along the streets in a daisy-chain style.

(ii) Each branch splits off to different houses.

There will be reduction in power

Reasons:

(i) Fiber attenuation : 0.2 dB/km.

(ii) Splitter losses : 0.5 dB per splitter.

(iii) Couple loss : 2 dB total.

1 dB at central office & 1 dB at the end user's location

From diagram

(i) House 13, 14 of 15 is at the end of the line.

(ii) They would pass through more splitters connecting 4 houses.

<u>Hub - design:</u>

- Each design has 1 x 4 splitter connecting 4 house to CO.
- Each fiber branch would be short
- House furtherest from their 1 X 4 splitter (4, 8, 12, 13) will have slightly lower power.
- However with longest fiber & splitter losses. likely make House 13 the lowest power

Let's calculate the recieved power.

Linear Design:

Fiber Distance from CO to house 13:

approx 16 km + 0.2 km + 0.2 km = 16.4 km.

Splitters : 3 No. (at CO & along line) $\rightarrow 0.5 + 0.5 + 0.5$
$= 1.5$ dB total.

Coupling loss is 2dB.

Total loss $= \alpha_{fiber} + \alpha_{splitters} + \alpha_{coupling}$

$= 0.2 \times (2 + 1.5 + 16.4) = 18.3$

$P_{Rx} = 0\,dBm - 18.3 \approx -18.3\,dBm \approx 0.0147\,mW.$

Hub design (eg for house 13)

- Fiber distance from CO to splitter : 16km

Fiber distance from splitter to house : 0.2km.

Splitter loss : 0.5 dB per 1X4 splitter.

Coupling : 2 dB

~~Splitter loss~~ : 6

$L_{fiber} = 0.2 \times (16\,km + 0.2)$

$= 3.24$ dB $L_{splitter}$

$L_{splitter} = 0.5$ dB

$L_{coupling} = 2$ dB

Total loss $L_{Total} = 3.24 + 0.5 + 2 = 5.74$ dB

Recieved power $P_{rx} = P_{tx} - L_{total} = 0 - 5.74$

≈ -5.74 dBm

$\boxed{P_{rx} \text{ in mW} = 0.266\ mW}$

2) a. Description and picture of the design.

Design description

1. Multiple Sources at the central Office (CO).

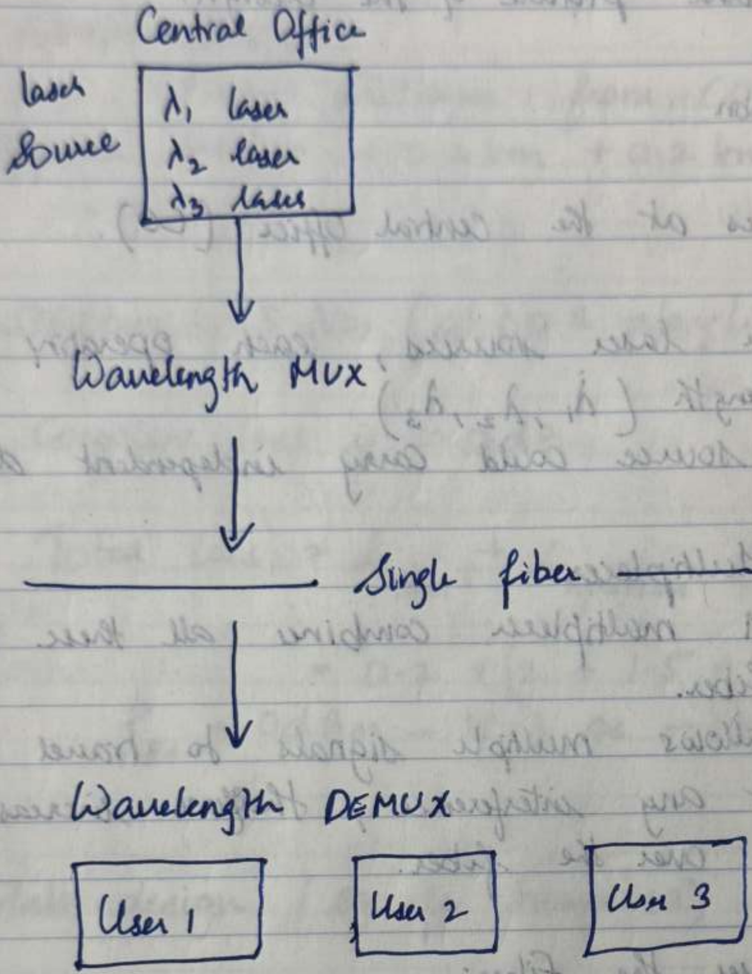
 - Multiple laser sources, each operating at a different wavelength (λ_1, λ_2, λ_3)
 - Each source could carry independent data streams

2. Wavelength Multiplexer.
 - A WDM multiplexer combines all these wavelength into a single fiber.
 - This allows multiple signals to travel wavelength together without any interference, therefore increasing the total bandwidth over the fiber.

3. Transmission over the fiber:
 - The single fiber carries the combined wavelengths through the neighbourhood.

4. Wavelength Demultiplexers (Demux)
 - At each location or at distribution points wavelength demultiplexers seperate the signals.
 - Each user location or at distribution points, ~~wavelength demultiplexers seperate the signals.~~

Design Diagram



- Optional splitters can be placed after the demux if multiple users share.

b) Advantages of this design over the original linear design.

Higher Bandwidth: Multiple wavelength all several data streams over a single fiber instead of sharing a single wavelength linearly.

Scalability: Adding a new user may only require only adding a new wavelength, rather than adding more fibers.

Instead of linear chain where distance from CO effects performance, all users share the fiber efficiently.

Each wavelength can act as a dedicated channel, improving performance & reducing interference.

c. Disadvantages over the original linear design.

(i) Cost: WDM / MUX / DEMUX, multiple laser sources & potentially more precise receivers increase equipment costs.
(ii) System design is more complex.
(iii) Adding splitters or long distances may require optical amplifiers to maintain signal strength
(iv) More components increase maintainence cost, efforts & any damage can be harder to find out.

3 Chromatic Dispersion

In optical fiber, different wavelengths travel at different speeds.

In our instance;

$\lambda_3 > \lambda_2 > \lambda_3 \Rightarrow$ so, λ_3 arrives first, λ_2 follows & finally λ_4.

Over long distances, this cause pulse broadening & overlapping, which can reduce signal quality.

Chirped fiber grab Bragg gratings.
· This would reflect different wavelengths at different points along a grating
: Longer wavelengths travel shorter optical path, shorter wavelengths travel longer paths, compensating for velocity differently.

3. Optical Phase Conjugators:
Convert wavelength differences ⇒ phase differences the evolve in a way that cancels dispersion downstream.

Step by step mechanism.

Let assume the system has DCF or CFBG after the main fiber
(1.) I/O : $\lambda_1 \lambda_2 \lambda_3$ pulse leave CO simultaneously but travel at diff velocities → λ_3 arrives first.
Compensation fiber/ grating.
· Insert a DCF with (−) dispersion.
· Slows down λ_3 relative to λ_2 & λ_1
· Speeds up λ_1 relative to λ_2 and λ_3.
or
Use a chirped Bragg grating.
· Reflects λ_1 later than λ_3 → effectively delay λ_3 less, λ_1 more → pulses realign.

3. <u>Output</u> : After the compensation component of all three wavelengths arrive more closely synchronized reducing inter - symbol interference.

A. Network (a) : Coupler based distribution.

Given:

$\boxed{2\times 2}$ coupler first: $\alpha = 0.5 \rightarrow$ splits i/p equally

Each arm goes into a $\boxed{1\times 2}$ coupler

$\alpha = 0.9 \rightarrow 90\%$ to one output, 10% to the other.

Inputs $\Rightarrow P_{\lambda_1}, P_{\lambda_2}, P_{\lambda_3}, P_{\lambda_4}$.

Outputs $\Rightarrow$ O1 O2 O3 O4

for 2×2 coupler $(\alpha = 0.5)$

- O/p powers $P_{\lambda i}$ ~~split~~ split equally into two arms.

for each i/p $P_{arm_1} = 0.5 P\lambda_i$, $P_{arm_2} = 0.5 P\lambda_i$

for 1×2 coupler $(\alpha = 0.9)$

90% of the input power goes to one O/p; 10% the other.

Apply to each arm from step 1.

$$P_{O\,maj} = 0.9 \times P_{arm_i} = 0.9 \times 0.5 P_{\lambda i} = 0.45 P_{\lambda i}$$
$$P_{O\,min} = 0.1 \times P_{arm} = 0.1 \times 0.5 P_{\lambda i} = 0.05 P\lambda_i$$

Assign Output:

O/p	Power	Wavelengths
O_1	$0.45 P\lambda_1 + 0.45 P\lambda_3$	λ_1, λ_3
O_2	$0.05 P\lambda_1 + 0.05 P\lambda_3$	λ_1, λ_3
O_3	$0.45 P\lambda_2 + 0.45 P\lambda_4$	λ_2, λ_4
O_4	$0.05 P\lambda_2 + 0.05 P\lambda_4$	λ_2, λ_4

Description

O_1 : Recieves the major split (90%) from the first arm, which carries λ_1 & λ_3. Thats each contributes to 0.5 of the original i/p power.
O_2 : Minor split (10%) from the first arm → 0.05 of each λ_1 & λ_3.
O_3 & O_4 : Same logic, but for the second arm carrying λ_2 & λ_4

Network (b) MZI - Based Power Distribution
MZI (Mach - Zehnder Interferometers)
MZI are wavelength dependent.
Transfer functions
· Solid line → O1/I1 & O2/I2, peaks at $\lambda 1, \lambda 3$ thru troughs at λ_2, λ_4.
· Dashed line → O1/I2 & O2/I1 phase shift
· This enables the wavelength selective routing.

Transfer functions
· Solid line → O1/I2 & O2/I1. peaks at $\lambda 2$ and $\lambda 4$.
Dashed line : O1/I1 & O2/I2. peaks at λ_1 and λ_3

Apply to 1st MZI layer.

$I_1 = P\lambda_1 + P\lambda_2$
$I_2 = P\lambda_3 + P\lambda_4$

O/p = O1a, O2a

O1a receives:

$\cdot$ I1 → O1 (dashed line) → peaks at λ_1, λ_3 → from I1 → High λ_2 → low.

$\cdot$ I2 → O1 (solid line) → peaks at $\lambda_2 \lambda_4$ → λ_4 from I_2 → High, λ_3 → low.

O1a ~~re~~ carries mostly λ_2 and λ_4

O2a receives:

$\cdot$ I1 → O_2 (solid line) → peaks at λ_2, λ_4 → λ_2 from I_1 → High λ_1 → low.

$\cdot$ I_2 → O_2 (dashed line) → Peaks at λ_1, λ_3 → λ_3 from I_2 → High, A_4 → low.

Apply to 2nd MZI layer.

$\cdot$ O1a → next MZI: → O1b, O2b.

Output	Wavelengths
O1a	$\lambda_1 \lambda_4$
O2a	$\lambda_2 \lambda_3$.

These now feed in 2nd layer MZI

$\cdot$ O1a → MZI → O1b, O2b

$\cdot$ O2a → MZI → O1C, O2C

Step 1: Apply transfer function to O1a → O1b, O2b
· I/p to this MZI : $\lambda 1$ and $\lambda 4$

Wavelength	O1b	O2b
λ_1 (from O1a)	Dashed line: O1 ⇒ Peak $1 \times P\lambda_1$	Dashed line → O_2 = trough $0 \times P\lambda_1$
λ_4 (from O1a)	O1 = trough → $0 \times P\lambda_4$	O_2 = Peak ⇒ $1 \times P\lambda_4$.

Results second layer:

Output	Wavelengths & relative power
O1b	$\lambda 1$ (high), $\lambda 4$ (low)
O2b	λ_1 (low), λ_4 (high)

Therefore, λ_1 mostly goes to O1b, λ_4 mostly goes O_2b.

Apply transfer function to O_2 → $O1_c$, $O2_c$

Inputs to this MZI: λ_2 & λ_3.

Wavelength	O12c	O2c
$\lambda 2$ (from O2a)	Solid line ↓ O1 = trough ↓ $0 \times P\lambda_2$	Solid line ↓ O2 = Peak ↓ $1 \times P\lambda_2$
λ_3 (from O2a)	Dashed line ↓ O_1 = Peak ↓ $1 \times P\lambda_3$	Dashed line ↓ O_2 = trough ↓ $0 \times P\lambda_3$

Results for second layer: (O1c, O2c)

Output	Wavelengths & Relative Power
$O1_c$	λ_3 (high), λ_2 (low)
$O2_c$	λ_2 (high), λ_3 (low)

So λ_3 mostly goes to $O1_c$, λ_2 mostly to $O2_c$.

Summary of Second layer Outputs.

Output	Dominant Wavelength	Notes.
O1b	λ_1	λ_4 minimal
O2b	λ_4	λ_1 minimal
O1c	λ_3	λ_2 minimal
O2c	λ_2	λ_3 minimal

5. Lasers Vs LED.

1. Similarities both are semiconductor light sources.
   · Both generate light via electron-hole recombination in a semiconductor.
   · Both are commonly used in optical communication systems.
2. Convert electrical energy into optical energy.
   · Electrical current injected into the device produces photons (light)
   · Both can be coupled into optical fibers for transmission.

Differences

Feature	LED	Laser.
Principle of operation	Spontaneous emission → photons emitted randomly	simulated emission → Coherent photons amplified in a cavity.
Optical output	Incoherent, broad spectrum, low intensity	Coherent, narrow linewidth, high intensity
Directionality	Emitted in many directions less collimated	Highly directional low divergence beam.
Threshold	No threshold	Requires threshold current to start lasing.
Modulation speed	Moderate (~100s of MHz)	High (~GHz or more)

b) Why high-speed detectors need very small detecting area.

1. Small area → low capacitance:
   - photodiode capacitance $C \propto$ area
   - Bandwidth $f \approx \dfrac{1}{2\pi RC}$
   - Small area → smaller C → higher bandwidth → faster response

2. Reduced transit time effects:

· Photogenerated carriers travel faster across small junctions → shorter delay → faster detection.

3. Better high-frequency response

· Large area slow down the rise/fall time → limits data rate

· Small-area detectors maintain clean, sharp electrical signals for Gb/s operation.

6. a) What determines the wavelength of a laser diode.

1) Semiconductor bandgap (E_g)

· Energy of photons: $E = h\nu = hc/\lambda$

· Photon energy ≈ bandgap energy → Wavelength.

$$\lambda \approx \frac{hc}{E_g}$$

· Different semiconductor materials (GaAs, InP, GaN, etc) produce different wavelengths.

2) Active region design / cavity structure.

· Fabry-Perot cavity length → only certain longitudinal modes can resonate → slightly affects wavelength selection.

· Refractive index of the material → Change effective cavity length → shifts wavelengths slightly.

3) Temperature: Bandgap decreases with increase in temperature → wavelength increases slightly (redshift).

6b) Basic p-in laser diode vs Quantum Well (QW) laser diode.

Feature	p-i-n laser Diode	Quantum Well (QW) laser diode
Active layer	Bulk intrinsic semiconductor	This quantum well layer.
Carrier Confinement	3D Bulk material	2D confinement → electrons & holes confined in the well.
Emission wavelength control	Bandgap of bulk ~~Relatively~~ material ~~high~~	Lower threshold
Threshold current	High	Low
Modulation speed	Moderate	High speed
O/p efficiency	Lower	High.

7. 1) Fabry - Perot resonance condition

The resonant wavelengths of a FP cavity are given by:

$$\lambda_m = \frac{2nL}{m}$$

where: n = refractive index of the cavity
L = Cavity length
m = mode number (integer)
λ_m = resonant wavelength.

We are designing the cavity to transmit only one channel, so the free spectral range (FSR) of the FP cavity should match the channel spacing.

2.) Free spectral range (FSR)

The FSR is the wavelength seperation between consequtive resonance:

$$\Delta\lambda_{FSR} \approx \frac{\lambda^2}{2nL}$$

$\lambda = 1.552\ \mu m$
$n = 3.25$
$\Delta\lambda = 2nm = 0.002\mu m$ (channel ~~packing~~ space)

Set FSR = channel spacing so only one channel resonates:

$$\Delta\lambda_{FSR} = \frac{\lambda^2}{2nL} \Rightarrow L = \frac{\lambda^2}{2n\Delta\lambda}$$

3. Substitute

$$\lambda = 1.552\,\mu m, \quad n = 3.25, \quad \Delta\lambda = 0.002\,\mu m.$$

$$L = \frac{(1.552)^2}{2 \times 3.25 \times 0.002\,\mu m}$$

$(1.552)^2 = 2.409\,\mu m^2$

Den $2 \times 3.25 \times 0.002 = 0.013$.

Div $2.409 / 0.013 = 185.3$

$$\boxed{L = 185.3\,\mu m}$$

8. Given:

absorption co-efficient $\alpha = 5000\ cm^{-1}$

Wavelength $\lambda = 1500\,nm$.

absorber length $L = 10\,\mu m = 1 \times 10^{-3}\ cm$.

Input optical power: $P_{in} = 1\,\mu W$

Desired Voltage $V = 0.5V$

(a) Calculate responsivity.

Optical absorption. The fraction of light absorped in the detector

$$\eta = 1 - e^{-\alpha L}$$

- $\alpha = 5000\ cm^{-1}$
- $L = 1 \times 10^{-3}\ cm$

$$\eta = 1 - e^{-5000 \times 0.001} = 1 - e^{-5}.$$

$e^{-5} \approx 0.0067$

$$\eta \approx 1 - 0.0067 \approx 0.9933 \approx 99.3\%$$

Almost all light is absorbed

2. Responsitivity formula.

$R\ (A/W)$

$$R = \eta \frac{q}{h\nu} = \eta \frac{q\lambda}{hc}$$

Where $q = 1.602 \times 10^{-19}$ C
$h = 6.626 \times 10^{-34}$ J
$c = 3 \times 10^{8}$ m/s.
$\lambda = 1500\text{ nm} \rightarrow 1.5 \times 10^{-6}$ m.

$$R = 0.9933 \times \frac{1.602 \times 10^{-19} \times 1.5 \times 10^{-6}}{6.626 \times 10^{-34} \times 3 \times 10^{8}}$$

Compute
Numerator $= 1.602 \times 10^{-19} \times 1.5 \times 10^{-6} \approx 2.403 \times 10^{-25}$.
Denominator $= 6.626 \times 10^{-34} \times 3 \times 10^{8} \approx 19.878 \times 10^{-26}$
$\approx 1.9878 \times 10^{-25}$

Div $:= 2.403 \times \cancel{10^{-25}} / 1.9878 \times \cancel{10^{-25}}$
$= 1.20$ A/W

$R \approx 1.20$ A/W

(b) Resistor to generate 0.5V

Photocurrent : $I_{ph} = R \cdot P_{in} = 1.20\text{A/W} \times 1 \times 10^{-6}\text{W}$
$= 1.2\ \mu A$

Voltage across resistor R: $V = I_{ph} \times R$

$$R = \frac{V}{I_{ph}} = \frac{0.5V}{1.2 \times 10^{-6}A}$$

$R \approx 4.166 \times 10^{5} \approx 4166\ k\Omega$

$\boxed{R = 4166 k\Omega}$

c) Why detector area affects speed

1. Capacitance effect

photodiode capacitance $C \propto$ area.

· RC time constant $\tau = R \cdot C$ limits the response speed.

2. Carrier transit time

· Large area → longer paths for carriers to reach electrodes → slow response